Unit 3 - Normalization

Database Management Systems | Semester V | Unit 3 of 5

1. Functional dependency

X -> Y holds when every pair of tuples agreeing on X also agrees on Y.

Armstrong's axioms: reflexivity, augmentation, transitivity.

2. Normal forms

1NF atomic attributes, no repeating groups.

2NF 1NF and no partial dependency on a composite key.

3NF 2NF and no transitive dependency.

BCNF every determinant is a candidate key.

3. Decomposition

Lossless join: the natural join of the parts reproduces the original.

Dependency preserving: every FD is enforceable inside one fragment.